

Brightlearn Data Analytics

Exercise 2 : SQL Foundations : Aggregate functions & Operators

01) SELECT DISTINCT department
FROM students ;

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department
IT
HR
Finance

02) SELECT department, AVG(age) AS avg_age
FROM students
GROUP BY department ;

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department	avg_age
IT	20,5
HR	22
Finance	23

03) SELECT department, COUNT(*) AS student_count
FROM students
GROUP BY department
HAVING COUNT(*) > 1;

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department	student_count
IT	2
HR	2

04) SELECT student_id, name, age, department
name ,
age ,
department
FROM students
WHERE age BETWEEN 21 AND 23

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student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

05) SELECT student_id,
name,
age,
department

FROM students
WHERE department IN ('IT', 'HR')
AND age > 21;

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student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

Part 2 : The courses table

06) SELECT department, SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM(credits) > 5;

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department	total_credits
IT	11

07) SELECT course_id,
course_name,
department,
credits
FROM courses
WHERE NOT credits = 4;

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course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

08) SELECT course_id,
course_name,
credits
FROM courses
ORDER BY credits DESC
LIMIT 3;

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course_id	course name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

Part 3: The enrollments Table

09) SELECT MAX(grade) AS max-grade,
MIN(grade) AS min-grade,
AVG(grade) AS avg-grade
FROM enrollments;

max-grade	min-grade	avg-grade
90	78	84,6

10) SELECT course_id, COUNT(*) AS enrollment_count
FROM enrollments
GROUP BY course_id;

course_id	enrollment count
101	1
102	1
103	1
104	1
105	1

Part 4: The salaries table

11) SELECT department,
SUM(salary) AS total_salary,
SUM(bonus) AS total_bonus
FROM salaries
GROUP BY department;

department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7 500
Finance	70 000	6 000

12) SELECT department
AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55 000;

department	avg_salary
IT	61 000
Finance	70 000

13) SELECT employee_id, name, salary, bonus,
salary + bonus AS total_compensation
FROM salaries
WHERE salary + bonus > 60 000;

employee_id	name	salary	bonus	total_compensation
1	Tom	60 000	5 000	65 000
3	Spike	70 000	6 000	76 000
4	Tyke	62 000	5 500	67 500

Part 5: The projects Table

14) SELECT department,
 SUM(budget) AS total_budget,
 AVG(budget) AS avg_budget
 FROM projects
 GROUP BY department
 HAVING AVG(budget) > 70 000;

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department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

15) SELECT project_id,
 project_name,
 department,
 budget
 FROM projects
 WHERE budget BETWEEN 50 000 AND 120 000
 AND department NOT IN('Marketing');

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project_id	project_name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
3	HR Portal	HR	50 000